Finite-Dimensional Irreducible Representations of sl2

An End-to-End Meta-Math Walkthrough

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Abstract

Raising must eventually stop because it creates distinct weights. Lowering from a highest-weight vector then forces the highest weight to be a nonnegative integer. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RT-05 / Representation Theory and Lie Theory / Report v1

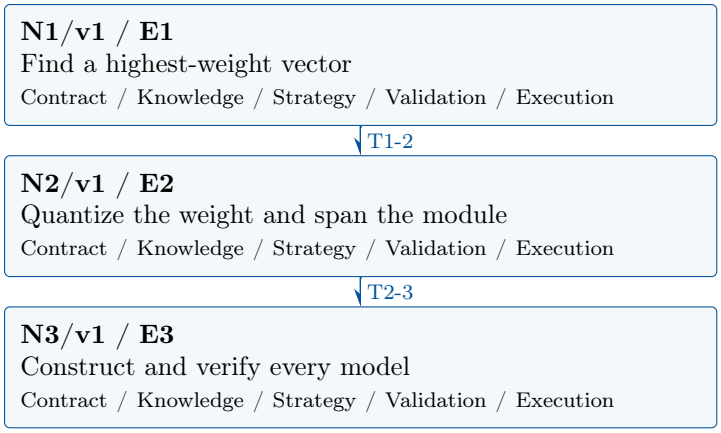
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let* $\mathfrak{sl}_2(\mathbb{C})$ *have generators* e, f, h *satisfying* $[h, e] = 2e$, $[h, f] = -2f$, *and* $[e, f] = h$. *Every nonzero finite-dimensional irreducible representation is, up to isomorphism, exactly one* V_n *for an integer* $n \geq 0$. *It has basis* $v_0, \ldots, v_n$ *with* $hv_j = (n - 2j)v_j$, $fv_j = v_{j+1}$ *(and* $fv_n = 0$*), and* $ev_j = j(n - j + 1)v_{j-1}$ *(and* $ev_0 = 0$*). In particular* $\dim V_n = n + 1$.

A. Foundations and permitted imports

Use eigenvalue existence over $\mathbb{C}$, linear independence of eigenvectors for distinct eigenvalues, the displayed Lie brackets, and polynomial interpolation on finitely many distinct scalars. Irreducibility is assumed for classification; this project does not prove that every finite-dimensional representation is completely reducible.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There is $v_0 \neq 0$ with $ev_0 = 0$ and $hv_0 = \lambda v_0$ for some $\lambda \in \mathbb{C}$.

Receiving activity. N2.1

Compatibility. E1/v1 and irreducibility of V; write $v_j = f^j v_0$. Check: The coefficient equation at $v_{n+1} = 0$ uses $v_n \neq 0$ and characteristic zero.

T2-3: N2/E2/v1 to N3. $\lambda = n \in \mathbb{Z}_{\geq 0}$ and $v_0, \ldots, v_n$ form a basis of V with the stated actions.

Receiving activity. N3.3

Compatibility. An arbitrary integer $n \geq 0$ and the formulas in the theorem. Check: Existence and irreducibility of the proposed models must be established in addition to the classification of a given module.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Find a highest-weight vector

Object and role. Start with an h-eigenvector and raise it until the last nonzero term. **Local obligation:** There is $v_0 \neq 0$ with $ev_0 = 0$ and $hv_0 = \lambda v_0$ for some $\lambda \in \mathbb{C}$.

Inputs. A nonzero finite-dimensional representation V.

C — Contract

Publish E1/v1 only after establishing this obligation: There is $v_0 \neq 0$ with $ev_0 = 0$ and $hv_0 = \lambda v_0$ for some $\lambda \in \mathbb{C}$.

K — Knowledge

A nonzero finite-dimensional representation V. Retain the stated assumptions and import versions.

S — Strategy

Start with an h-eigenvector and raise it until the last nonzero term.

V — Validation

The raising chain has distinct weights; finite-dimensionality, not a preassumed diagonalization of all of h, forces termination.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Choose $w \neq 0$ with $hw = \alpha w$. The relation $he = eh + 2e$ gives $h(e^j w) = (\alpha + 2j)e^j w$ by induction. Nonzero members of this sequence are eigenvectors with distinct eigenvalues and hence linearly independent.

N1.2 They cannot all be nonzero in finite dimension. Choose the last nonzero one before the first zero, and name it v_0. It is killed by e and has an h-eigenvalue λ, as required.

Export E1/v1. There is $v_0 \neq 0$ with $ev_0 = 0$ and $hv_0 = \lambda v_0$ for some $\lambda \in \mathbb{C}$.

Acceptance gate. The raising chain has distinct weights; finite-dimensionality, not a preassumed diagonalization of all of h, forces termination.

Targeted repair. If h is declared diagonalizable at the outset, replace that unproved claim by one eigenvector and the finite raising chain. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Quantize the weight and span the module

Object and role. Commute e through powers of f and inspect the first vanishing lowered vector.
Local obligation: $\lambda = n \in \mathbb{Z}_{\geq 0}$ and $v_0, \ldots, v_n$ form a basis of V with the stated actions.

Inputs. E1/v1 and irreducibility of V; write $v_j = f^j v_0$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\lambda = n \in \mathbb{Z}_{\geq 0}$ and $v_0, \ldots, v_n$ form a basis of V with the stated actions.

K — Knowledge

E1/v1 and irreducibility of V; write $v_j = f^j v_0$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Commute e through powers of f and inspect the first vanishing lowered vector.

V — Validation

The coefficient equation at $v_{n+1} = 0$ uses $v_n \neq 0$ and characteristic zero.

E — Execution

Carry out activities N2.1–N2.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Inductively, $hv_j = (\lambda - 2j)v_j$. Also $ev_0 = 0$ and $ev_{j+1} = efv_j = fev_j + hv_j$, so induction gives $ev_j = j(\lambda - j + 1)v_{j-1}$. These identities use only the brackets and remain true until and including the first zero vector.

N2.2 The nonzero lowered vectors have distinct weights, so finite dimension gives a largest n with $v_n \neq 0$ and $v_{n+1} = 0$. Applying the formula at $n + 1$ yields $0 = (n + 1)(\lambda - n)v_n$. Therefore $\lambda = n$.

N2.3 The span of $v_0, \ldots, v_n$ is invariant under e, f, h by these formulas and is nonzero. Irreducibility forces it to equal V. Distinct weights give linear independence, so these vectors are a basis and all claimed coefficients follow.

Export E2/v1. $\lambda = n \in \mathbb{Z}_{\geq 0}$ and $v_0, \ldots, v_n$ form a basis of V with the stated actions.

Acceptance gate. The coefficient equation at $v_{n+1} = 0$ uses $v_n \neq 0$ and characteristic zero.

Targeted repair. If termination of lowering is asserted without justification, use the distinct eigenvalues; then apply the formula at the first zero, not merely at v_n. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Construct and verify every model

Object and role. Check the brackets directly and recover the highest vector from any nonzero invariant subspace.
Local obligation: For each n, V_n exists and is irreducible; these exhaust all isomorphism classes without duplication.

Inputs. An arbitrary integer $n \geq 0$ and the formulas in the theorem. Incoming proof exports: E2/v1 at N3.3.

C — Contract

Publish E3/v1 only after establishing this obligation: For each n, V_n exists and is irreducible; these exhaust all isomorphism classes without duplication.

K — Knowledge

An arbitrary integer $n \geq 0$ and the formulas in the theorem. Incoming proof exports: E2/v1 at N3.3. Retain the stated assumptions and import versions.

S — Strategy

Check the brackets directly and recover the highest vector from any nonzero invariant subspace.

V — Validation

Existence and irreducibility of the proposed models must be established in addition to the classification of a given module.

E — Execution

Carry out activities N3.1–N3.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 Define three linear maps on the abstract basis by the stated formulas. The weight differences give $[h, e] = 2e$ and $[h, f] = -2f$. Direct subtraction gives $[e, f]v_j = ((j+1)(n-j) - j(n-j+1))v_j = (n-2j)v_j = hv_j$, with the same result at $j = 0, n$ using zero boundary terms. Thus they define a representation.

N3.2 Let U be a nonzero invariant subspace. A nonzero vector of U has a nonzero component along some v_j. Because the h-eigenvalues are distinct, a polynomial in h that equals one at that weight and zero at the other weights extracts a nonzero multiple of v_j within U. Applying e^j gives a nonzero multiple of v_0, since all coefficients $r(n-r+1)$ for $1 \leq r \leq j$ are nonzero. Repeated f then yields every basis vector. Hence $U = V_n$.

N3.3 N2 identifies any irreducible module with its corresponding model by sending the displayed basis to the abstract one. Different n give different dimensions, so the models are pairwise nonisomorphic. This proves existence, uniqueness, and exhaustion.

Export E3/v1. For each n, V_n exists and is irreducible; these exhaust all isomorphism classes without duplication.

Acceptance gate. Existence and irreducibility of the proposed models must be established in addition to the classification of a given module.

Targeted repair. If only the possible highest weights have been listed, add bracket verification and the invariant-subspace argument to establish that every listed model really exists and is irreducible. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Find a highest-weight vector

1. **N1.1.** Choose $w \neq 0$ with $hw = \alpha w$. The relation $he = eh + 2e$ gives $h(e^j w) = (\alpha + 2j)e^j w$ by induction. Nonzero members of this sequence are eigenvectors with distinct eigenvalues and hence linearly independent.

2. **N1.2.** They cannot all be nonzero in finite dimension. Choose the last nonzero one before the first zero, and name it v_0. It is killed by e and has an h-eigenvalue λ, as required.

B. N2 — Quantize the weight and span the module

1. **N2.1.** Inductively, $hv_j = (\lambda - 2j)v_j$. Also $ev_0 = 0$ and $ev_{j+1} = efv_j = fev_j + hv_j$, so induction gives $ev_j = j(\lambda - j + 1)v_{j-1}$. These identities use only the brackets and remain true until and including the first zero vector.

2. **N2.2.** The nonzero lowered vectors have distinct weights, so finite dimension gives a largest n with $v_n \neq 0$ and $v_{n+1} = 0$. Applying the formula at $n + 1$ yields $0 = (n + 1)(\lambda - n)v_n$. Therefore $\lambda = n$.

3. **N2.3.** The span of $v_0, \ldots, v_n$ is invariant under e, f, h by these formulas and is nonzero. Irreducibility forces it to equal V. Distinct weights give linear independence, so these vectors are a basis and all claimed coefficients follow.

C. N3 — Construct and verify every model

1. **N3.1.** Define three linear maps on the abstract basis by the stated formulas. The weight differences give $[h, e] = 2e$ and $[h, f] = -2f$. Direct subtraction gives $[e, f]v_j = ((j + 1)(n - j) - j(n - j + 1))v_j = (n - 2j)v_j = hv_j$, with the same result at $j = 0, n$ using zero boundary terms. Thus they define a representation.

2. **N3.2.** Let U be a nonzero invariant subspace. A nonzero vector of U has a nonzero component along some v_j. Because the h-eigenvalues are distinct, a polynomial in h that equals one at that weight and zero at the other weights extracts a nonzero multiple of v_j within U. Applying e^j gives a nonzero multiple of v_0, since all coefficients $r(n - r + 1)$ for $1 \leq r \leq j$ are nonzero. Repeated f then yields every basis vector. Hence $U = V_n$.

3. **N3.3.** N2 identifies any irreducible module with its corresponding model by sending the displayed basis to the abstract one. Different n give different dimensions, so the models are pairwise nonisomorphic. This proves existence, uniqueness, and exhaustion.

□

V. WORKED ORCHESTRATION

For $n = 2$, the weights are $2, 0, -2$. The lowering chain is $v_0 \mapsto v_1 \mapsto v_2 \mapsto 0$, and $ev_1 = 2v_0$, $ev_2 = 2v_1$. At the middle vector, $efv_1 = 2v_1$ and $fev_1 = 2v_1$, so their difference equals $hv_1 = 0$. At v_0, the difference is $2v_0 = hv_0$.

A. Boundary and hypothesis audit

Irreducibility is essential for the conclusion that one chain spans the entire module. The direct sum $V_0 \oplus V_0$ has dimension two but $h = e = f = 0$; it is not the irreducible V_1, whose h weights are $1, -1$.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 conclusion: a highest-weight chain spans every finite-dimensional representation. **Validation witness:** $V_0 \oplus V_0$, where any generated chain is just one line. **Diagnosis:** the passage from a nonzero invariant span to all of V silently used irreducibility. **Repair:** invoke the stated irreducibility hypothesis precisely at N2.3. No general complete-reducibility claim is added; recheck the basis and dimension exports.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Develop the separate complete-reducibility theorem and then tensor-product decomposition rules, keeping them distinct from this irreducible classification.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.